



Working with Medical Volumes

Often, you need to do more than just view your data. For example, you may need to extract specific slices for future processing.

Navigate to the Module 2 folder and open the **WorkingWithMedicalVolumes.mlx** file to import a medical volume, view the metadata, and extract slices from the volume.

We recommend working with the code in MATLAB to gain experience and experiment and practice. A static version of the code is provided below. The static version does not include the interactive tools needed to run the code.

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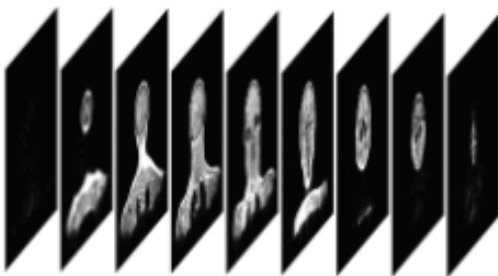
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Overview

You've seen how to visualize 3D medical images in MATLAB using the Medical Image Labeler App. In this script, you'll learn how to import medical volumes and use an alternative approach to visualize the image quickly. You'll also extract individual slices of data for further analysis.

3D medical images can be thought of as a stack of 2D images sampled along a third dimension. The stack of images is stored as a 3D array. The elements of the array are often referred to as 3D pixels called voxels. The value at each voxel represents the intensity recorded for that location.



Import 3D Medical Images

Use the `medicalvolume` function to import a 3D image. When the volume is saved as a directory of images, the input must be the folder containing the volume files.

Use the button below to select one of the MRI folders provided with the course files.

```
volumeFolder = '';  
mri = medicalvolume(volumeFolder)
```

Similar to a `medicalImage` object, a `medicalVolume` contains the image data along with important meta-data about the image.

- * `voxels` is the 3D array storing the image data

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